Syllabus

MAT 5173 — Algebra I Fall 2026 · UT San Antonio

Texts and Materials

Primary text

- **Ash** — Ash, *Abstract Algebra: The Basic Graduate Year*, 2002

Secondary reading

- **LiZh** — Li & Zhao, *Introduction to Abstract Algebra*, Les Ulis: EDP Sciences, 2022. ISBN 978-2-7598-2915-6. DOI: 10.1051/978-2-7598-2916-3. Gentler (undergraduate-level) discussion of many topics in Ash. (Primary text for MAT 4233's.)
- **CLO** — Cox, Little & O'Shea, *Ideals, Varieties, and Algorithms*, 5th ed., Springer, 2025. ISBN 978-3-031-91840-7. DOI: 10.1007/978-3-031-91841-4. Algebraic geometry text; supplements Ash Ch. 8.

Supplementary reading

- **Shafarevich** — Shafarevich, *Basic Algebraic Geometry 1*, 3rd ed., Springer, 2013. ISBN 978-3-642-37955-0. DOI: 10.1007/978-3-642-37956-7
- **Rosebrock** — Rosebrock, *Visual Group Theory*, Springer, 2024. ISBN 978-3-662-69364-3. DOI: 10.1007/978-3-662-69365-0
- **Beardon** — Beardon, *Algebra and Geometry*, Cambridge University Press, 2005. ISBN 978-0-521-81362-4. DOI: 10.1017/CBO9780511800436

Course Information

Catalog Entry

MAT 5173 Algebra I

The opportunity for development of basic theory of algebraic structures. Areas of study may include monoids, semigroups, groups, isomorphism theorems, free groups, group extensions and group actions, Sylow theorems, group chains and composition series, nilpotent and solvable groups, cohomology of groups.

Prerequisites

MAT 4233 or consent of instructor.

Credit Hours

3

Course Modality

Traditional in-person

Class Meetings

Duration	Wed 19 Aug – Thu 3 Dec

Campus	Main Campus
Location	MH 3.03.10
Time	Mon Wed 7:30PM–8:45PM

Learning Objectives

By the end of this course you should be able to:

- Define the notions of group, cyclic group, subgroup, normal subgroup, homomorphism, isomorphism, permutation group.
- Describe and analyze examples of groups, emphasizing groups of geometric transformations.
- Define group actions, emphasizing geometric actions.
- Define the notions of ring, subring, domain, field, ideal, homomorphism, isomorphism.
- Define rings of polynomials over a field, the notion of degree, and irreducible factors.
- Explore connections of the abstract theory of rings, polynomials and ideals with the geometry of algebraic varieties.
- Communicate concepts in writing and debate technical arguments orally.

About the Instructor

Eduardo Dueñez, Associate Professor of Mathematics.

Department	Mathematics
Office	FLN 4.01.11
Student hours	Mon 5:30PM – 6:30PM & Tue 1:30PM – 2:30PM
Email	eduardo.duenez@utsa.edu
Homepage	https://supernumero.us/about

Email is the preferred method of communication.

Assessment and Grading

Weight of course activities

Component	Weight	
Q&A sessions	40%	4 sessions, 10% each; none dropped
Problem sets	15%	4 sets, 3.75% each; graded for completeness
Midterm Exam 1	20%	In class, Mon 5 Oct
Midterm Exam 2	25%	Wed 2 Dec in class

Q&A sessions

Four class meetings throughout the semester (roughly every other Wednesday except when adjacent to an exam) will be *Q&A sessions* in which each student will be asked to explain and defend their solution to a homework problem submitted as part of a written assignment. You will only be asked questions related to an assigned problem and about the solution you submitted.

Each Q&A session is scored 0–3:

- **3** Correct explanation to a correct solution, demonstrating full understanding of the concepts and all aspects of the solution submitted.
- **2** A completely correct solution submitted is explained to an incomplete degree; **Or**: an unfinished solution, with a thorough and articulate account of what you tried, explaining where and why you got stuck while demonstrating knowledge of the background and reading necessary to solve the problem.
- **1** Correct solution inadequately explained revealing little or no understanding; *or*: partial solution accompanied by an incomplete but partly correct oral explanation.
- **0** Nothing submitted; *or*: work submitted (even if fully correct) is backed by no cogent explanation nor knowledge of concepts involved.

A correct answer you cannot explain is worth less than an explainable honest failure.

Note: A low Q&A average caps the course grade, whatever the other components come to:

- a Q&A average **below 1.5** caps the course grade at **C**;
- a Q&A average **below 1.0** caps the course grade at **D**.

(An average below 1.0 is of course also below 1.5; in that case the 'D' cap is the one that applies.)

Problem sets

Four sets, each with five or six theoretical problems. **Maximum four pages.**

Computational (SageMath) material is supplementary and entirely optional: it is never part of a problem set or a Q&A session, and it is not graded.

Each item will be scored *only* based on the **degree of completion** (not on correctness): **2** for a nominally complete solution, **1** for a clearly partial one, **0** for nothing submitted or for obvious filler with no substance.

Note: Any submission containing non-human-readable content, stray characters, or artifacts revealing non-proofread AI output scores 0 for that item.

Use of Artificial Intelligence (AI)

AI assistance is allowed in preparing work for submission, as are any solution keys, collaboration, and internet sources. You are ultimately responsible for your submissions; any part of your work that has clearly not been proofread (e.g., includes non-human readable characters or LLM artifacts) automatically earns a zero grade. *Include one line noting what resources you used in your submission.*

This policy is by design, and it reflects how the course is graded: 85% of the grade comes from Q&A sessions and in-class exams where only what is truly understood and can be explained counts. Assignments submitted but not written/understood by you result in failing grades in Q&A sessions and exams. Use AI to learn faster —not to prepare and turn in papers you cannot defend, blocking you from passing exams as well.

All audiovisual (or audio only, or video only) recordings of lectures are explicitly forbidden; all uses of AI agents or apps to capture, process, or analyze any lecture contents are **strictly forbidden**. The only *limited* exceptions are *explicitly* allowed recordings per official memorandum of the office of Student Disability Services. Under no circumstances is any processing or analyzing of such recordings allowed. Any sharing of confidential course materials with individuals who are not enrolled students in the course is a violation of FERPA privacy laws.

Exams

Both exams are traditional in-class (written and closed book). Exams are based **primarily on the assigned problem sets** —expect some context and phrasing to differ from the homework, plus a minority of other unseen problems. Memorizing solutions will not result in good exam grades; preparing by understanding the assigned problems will make the exams straightforward and earn high Q&A grades as well.

Attendance and missed work

No extensions on problem sets. A *single* missed Q&A session in the semester may be made up during office hours within seven days (no documentation is required). *No Q&A scores are dropped.*

Final grade ranges

Grade	Range	Grade	Range
A	[90%, 100%]	A-	[85%, 90%)
B+	[80%, 85%)	B	[76%, 80%)
B-	[72%, 76%)	C+	[69%, 72%)
C	[65%, 69%)	C-	[60%, 65%)
D	[50%, 60%)	F	[0%, 50%)

Time Commitment Expectations

Expect an average of about **9 hours per week**, and no fewer than 7.

- Attend class meetings (**3 hours per week**)
- Reading the assigned textbook sections (**2–3 hours per week**)
- Problem sets (**2–3 hours per week**)
- Office hours and study group (**1 hours per week**)

Additional Course Information

Each student should be intimately familiar with the contents of any work submitted for grading. The instructor has the right to request adequate **verbal explanation** of methodology and content for any submitted work. Credit will not be awarded when such an explanation is requested but not satisfactorily provided.

Course Schedule

Date	Topic
Wed 19 Aug	Lecture — Course launch; groups and subgroups
Mon 24 Aug	Lecture — Permutation groups; cyclic groups; the order of an element
Wed 26 Aug	Lecture — Cosets, normal subgroups and homomorphisms
Mon 31 Aug	Lecture — The isomorphism theorems
Wed 2 Sep	Lecture — Direct products
Wed 9 Sep	**Q&A** (**Due**: Set 1)
Mon 14 Sep	Lecture — Rings: basic definitions and properties
Wed 16 Sep	Lecture — Ideals, homomorphisms and quotient rings; the isomorphism theorems for rings

Date	Topic
Mon 21 Sep	Lecture — Maximal and prime ideals
Wed 23 Sep	Lecture — Polynomial rings; unique factorization
Mon 28 Sep	Lecture — Principal ideal domains and Euclidean domains; rings of fractions; irreducible polynomials
Wed 30 Sep	**Q&A** (**Due**: Set 2)
Mon 5 Oct	**Exam** — Midterm Exam 1 — Sets 1 and 2
Wed 7 Oct	Lecture — Field extensions; degree; simple extensions
Wed 14 Oct	Lecture — Splitting fields; algebraic closures
Mon 19 Oct	Lecture — Separability; normal extensions; finite fields
Wed 21 Oct	Lecture — Affine space and affine varieties
Mon 26 Oct	Lecture — V(I) and I(V): algebra ↔ geometry
Wed 28 Oct	Lecture — The coordinate ring; the algebra-geometry dictionary; Hilbert's Nullstellensatz
Mon 2 Nov	**Q&A** (**Due**: Set 3)
Wed 4 Nov	Lecture — The Hilbert basis theorem; Noetherian rings
Mon 9 Nov	Lecture — Plane curves: rational curves; nodes and cusps
Wed 11 Nov	Lecture — Fixed fields and Galois groups
Mon 16 Nov	Lecture — The fundamental theorem of Galois theory
Wed 18 Nov	Lecture — Survey: the unsolvability of the quintic
Mon 23 Nov	**Q&A** (**Due**: Set 4)
Mon 30 Nov	*Review* — Review — the exam pool
Wed 2 Dec	**Exam** — Midterm Exam 2 — Sets 3 and 4

Readings for each meeting are on the Weekly Schedule (also available as a printable PDF).

Department and University Policies

Ombudsperson

The ombudsperson is an advocate who investigates complaints about a specific section or instructor and attempts to resolve them through mediation. Comments or complaints sent to this person will be brought to the attention of department leadership, who will follow up. If you have questions about the logistics of your class, please contact the instructor.

Contact Ilse Rosales Martinez at ilse.rosalesmartinez@utsa.edu.

Video and Audio Recording

As the instructor of this course, I may record meetings and lessons. You are expected to follow appropriate University policies and maintain the security of passwords used to access recorded lectures. Recordings may not be published, reproduced, or shared with those not in the class. If the instructor or a UTSA office plans any other uses for the recordings, consent of the students identifiable in the recordings is required before such use unless an exception is allowed by law.

Syllabus Changes

The syllabus is subject to change at the instructor's discretion. Any changes or corrections to the course materials, assignment dates, or other updates will be communicated to students ahead of time. You are responsible for checking Canvas for corrections or updates to the syllabus.

Essential Student Information

- **Important:** Bookmark and visit the Common Syllabus Information webpage for resources about counseling services, transitory/minor medical issues, supplemental instruction, tutoring services, academic success coaching, sexual harassment and sexual misconduct, campus safety and emergency preparedness, and the Roadrunner Creed.
- For technical requirements, support, and resources, visit Academic Innovation's Student Technical Support page.
- UT San Antonio provides reasonable accommodations to students via Student Disability Services, or call (210) 458-4157.
- Your well-being is important. Support, including 24/7 mental health services, can be accessed from the Well-being at UT San Antonio site.
- Student Assistance Services offers confidential support and personalized guidance.
- The Roadrunner Pantry provides free access to groceries, toiletries, and other essentials.
- Students are responsible for ensuring their work is consistent with UTSA's standards for academic integrity. Review Section 203 of the Student Code of Conduct.
- Visit UT San Antonio Libraries and Museums for journals, research tutorials, and your department's librarian.